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Global Thematics | North America

Weighing In: Open-Weights Models & 3 States of the World

We believe the progression of open-weights models aids Jevons Paradox and AI adoption. Enterprises use open & closed-weights models today — benefitting in cost, customization, and control — but how this trends remains to be seen. We outline "3 States of the World" and highlight stock beneficiaries.

Key Takeaways

- Open-weights models have become a recent debate, with Kimi K3 / potential US gov't restrictions.
- We believe open-weights models are good for competition and can enable faster AI diffusion, supported by Jevons Paradox, and see a long runway for AI adoption.
- 60%+ of enterprises use open-weights models as part of their stack, largely isolated to specific use cases requiring speed, security, or frequent use.
- We outline 3 States of the World — where closed-weights models win, hybrid wins, and open-weights models win — and highlight key stock beneficiaries.
- Closed wins: Like closed model players/AVGO. Open wins: Like open model players/security/on-prem infra/edge. NVDA/power always win, hyperscalers with nuances.



Tech Diffusion

A Morgan Stanley Research
Key Theme of 2026

Open-weights models have become increasingly important to the AI narrative.

The open vs. closed-weights model debate has reached a fever pitch over the past weeks with 1) the release of Kimi K3, 2) potential US government restrictions of Chinese models, and 3) the "Open Weights and American AI Leadership" and Anthropic letters. Investors are concerned that compute and AI infrastructure demand will fall with the proliferation of smaller, more efficient open-weight Chinese models. We believe open-weights models are good for competition and can enable faster AI diffusion, supported by Jevons Paradox. This is important given we see a long runway for enterprise AI adoption (see math showing an average AI task provides a \$55 benefit vs. \$2-5 cost [here](#)). **The big question is: Who wins — open or closed-weights models — and what does the world look like in each scenario?**

Many enterprises are using both closed and open-weights models today.

According to a [McKinsey survey](#), 63% of surveyed companies use open-weights models today in combination with closed ones. We believe these open-weights deployments are largely isolated to smaller, specific use cases that require speed,

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security, or frequent calls — like coding or document parsing. Key benefits of open-weights models include no per-token costs, customization, and control over model hosting. Drawbacks include the cost of fine-tuning models in-house (requires talent and resources), security risks, liability attribution, and geopolitical risk. Potential bans of Chinese open-weights models in the US create business continuity risk, potentially causing US-based companies to stick with frontier model providers or US-based open-weights options. How the use of open-weights models evolves over the coming years is an open question.

We outline "Three States of the World" for open vs. closed-weights models and highlight key stock beneficiaries. In **Scenario 1 (closed models win)**, we see a small number of labs win. Cost per token falls, but not as fast as in an open-weights scenario due to the market oligopoly. In **Scenario 2 (hybrid)**, no single model architecture wins — closed frontier models dominate for complex workloads, while open-weights models handle high-volume, cost-sensitive or domain-specific tasks. AI deployment spans public/private cloud, on-prem, and edge. Orchestration and observability become critical, as enterprises run many models — both open and closed. In **Scenario 3 (open-weights wins)**, open-weight models converge with frontier-level performance and model prices fall more drastically, leading to greater enterprise adoption. Innovation shifts from pre-training toward fine-tuning, inference optimization, agents, tools and application-specific deployments. Smaller models imply more workloads move on-prem and to the edge, benefitting on-prem infrastructure and edge devices. Security software becomes an increasing winner due to on-prem / distributed workloads.

For Meta Marshall's thoughts on implications for security software/networking, see [link](#).

Exhibit 1: Stocks we like across our "3 States of the World" — where closed models win, hybrid wins, and open models win.

	Scenario 1 (Closed Models Win)	Scenario 2 (Hybrid)	Scenario 3 (Open Models Win)
Biggest Beneficiaries	Cloud Providers: GOOGL, AMZN Model Providers: GOOGL, META Security Software: PANW, CRWD, ZS, NTSK, OKTA Optical/Networking: ANET, LITE, COHR On-Site Power Providers: BE, INIO, SEI, WMB, LBRT Semis: NVDA, AVGO	Cloud Providers: AMZN, GOOGL, MSFT Infrastructure Software: DDOG, PLTR, APPN Security Software: PANW, CRWD, FTNT, ZS, NTSK, OKTA, SAIL, VRNS SaaS: SAP, NOW Optical/Networking: CSCO, FFIV On-Site Power Providers: BE, INIO, SEI, WMB, LBRT Semis: NVDA	Cloud Providers: MSFT Model Providers: MiniMax, Knowledge Atlas, BABA, Tencent Infrastructure Software: PLTR Security Software: PANW, CRWD, FTNT, OKTA, SAIL SaaS: SAP, NOW, SHOP On-Prem Infrastructure: DELL, HPE, NTAP, P Edge Devices: DELL, HPQ, AAPL VARs/Distributors: SNX, INGM, CDW On-Site Power Providers: BE, INIO, SEI, WMB, LBRT Semis: NVDA
Other Beneficiaries	Cloud Providers: MSFT Neoclouds: CRWV, NBIS Security Software: SAIL	Model Providers: MiniMax, Knowledge Atlas, BABA, Tencent Neoclouds: CRWV, NBIS SaaS: SHOP, WDAY, CRM, INTU On-Prem Infrastructure: DELL, HPE, NTAP, P Edge Devices: DELL, HPQ, AAPL VARs/Distributors: SNX, INGM, CDW	Cloud Providers: AMZN, GOOGL Neoclouds: CRWV, NBIS SaaS: WDAY, CRM, INTU

Source: Morgan Stanley Research

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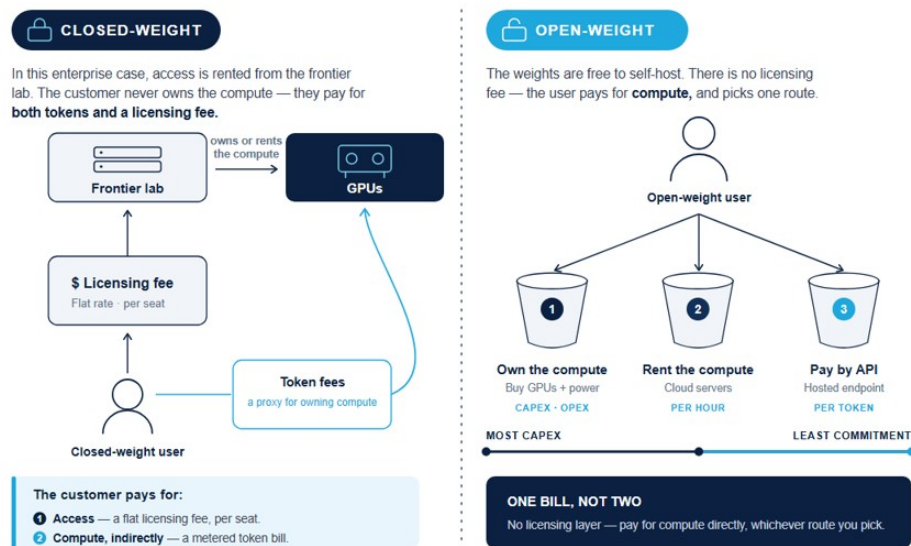
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The Debate on Open-Weights Models

Overview of Open-Weights Models

We believe that the market fundamentally misunderstands the role and functionality of open-weights models. Historically, open-weights models tend to be smaller (in number of total parameters), faster, and more cost-efficient; as such, the wider audience came to expect these models to be of "lower quality" or "light" versions of the higher-parameter frontier models (known as closed-weight models). The difference between open-weights (or open-weight) models and closed weight models has nothing to do with the number of active parameters, overall cost, or efficiency. The distinction between the two relates to how users pay and/or interact with the models: open-source models are **1)** free to access, as opposed to gated by subscriptions, **2)** able to be trained and tuned (weights are "open" or able to be changed), as opposed to be the same for every user (weights are "closed"), and **3)** give ultimate deployment flexibility, as it can be run on-prem, in the cloud, or through an API (either model creator's API, hosted service like AWS Bedrock, or self-hosted API; outlined in [Exhibit 2](#)). Closed-weight models, alternately, can either be accessed directly from a model provider (the user pays a licensing fee, like a subscription, for access, and a token cost that serves as a proxy for GPU usage) or through a cloud provider should PII be a concern.

Exhibit 2: Enterprise Closed Weight Users Incur Both Licensing and Token Costs While Open-Weight Users Pay for Their Usage in Three Main Ways



Source: Morgan Stanley Research

So: what are they? Put simply: open-source models are simply models that can be adapted to a specific use case (provided a user has sufficient time, money, and compute) while closed weight models are the same for every user. Open-weight or closed-weight is not necessarily a permanent state, rather describing the permissions of a user's access. The highest-performing models, like those produced by the frontier labs, are closed weight because of the incredible monetary costs with producing them. These companies cannot afford to give these models away until they release their next generation of

models. It is at this point that the former "best" models become a part of a "freemium" model to encourage non-paying users to adopt the technology while gating them from the best available product.

What is the difference between open-weights and open source models? Open-weight models release their trained parameters but may withhold training data, code, or unrestricted licensing. Examples include Meta's Llama, Google's Gemma, Qwen, DeepSeek, and Kimi (under a strict definition). Open-source models release the broader stack under open licenses, often including weights, code, training guidelines, and data details. Examples include AI2's OLMo.

Distribution of open-source models is another key debate. Releasing a strong open-source model wins market share and a customer base while better enabling the spread of technology (which benefits all players). Some labs release open-source models — or progressions in model architecture — in alignment with academic research, where progressions in the field benefit all players. Others hypothesize that the release of models like K3 or Qwen 3.8 are deliberate attempts to undermine frontier labs' credibility; prior to models like Qwen, frontier labs showed clear advantages in model efficiency. Until the release of K3, closed-weight frontier models were seen as being clear leaders in accuracy/benchmarking. This dynamic is a part of nascent market where we see multiple labs jockey for position, competing on model metrics like price, accuracy, benchmarking, speed, overall efficiency, and safety.

What Does Enterprise Adoption of Open-Weights Models Look Like Today?

Open-weights models are heavily used as part of companies' AI model tech stacks.

According to a [McKinsey survey](#) of 700 tech leaders across 41 countries, **63% of respondents use open source models**, often alongside closed options like the frontier labs. Recent stats indicate US companies [have routed more than 30%](#) of their OpenRouter tokens to Chinese open models every week from February to July 2026 (although this likely skews towards startups than enterprises). While we still think open-weights models account for the minority of LLM usage at enterprises, this shows that companies are more aggressively experimenting with an open/closed hybrid approach: it's important to understand how the mix between open and closed-weights models could shift over time.

We'd cite a few key benefits of open-weights models to enterprises:

1. **Eliminate Token Costs (but this doesn't mean it's free).** Given customers of open-source pay *nothing* in upfront costs (most, if not all, open-weights models are free), they are then free to call it as many times as they want at zero cost. Please see [Exhibit 2](#) for an explanation of potential costs of open- and closed-weight models. Open-weight models still require hosting costs or physical compute, plus the cost of training, so it's not accurate to call them free on a TCO basis. However, open-weights models could help lower opex burdens and mitigate ballooning per-token inference costs — particularly useful for use cases with high query volume. ML Leader at Shopify (Kshetrajna Raghavan) noted: "*When you need to run billions of multi-modal inferences a day, [closed model] per-token*

charges aren't sustainable." A study from [MIT](#) noted that a shift from closed to open models could reduce average prices by 70% and save consumers ~\$25B per year. We would caveat that 1) this analysis was done in December 2025; 2) not all open-weights models are cheaper on a per-token basis, as we've seen with Kimi K3; and 3) there are other costs associated with open-weights models (more below). However, we think this is illustrative of potential variable cost savings for enterprises.

2. **Customization.** Fine-tuning a model to industry standards or company's internal data (like customer support logs, internal documents, domain-specific tasks) can be highly effective from a performance standpoint, which can be key for highly specialized or repetitive tasks on the same data but costly to execute depending on the scale of training/fine-tuning.
3. **Control Over Hosting.** With the increasing importance of data sovereignty and localization rules, open-weights models give companies control over where to host their models (on-prem, private cloud, etc.). A survey from [McKinsey](#) showed that 40% of enterprise leaders like open-weights models for this reason. This is particularly beneficial for highly regulated industries with local hosting requirements.

But there are also a few drawbacks:

1. **Cost / Time of Fine-Tuning Models to Stay at the Leading Edge.** Fine-tuning an open-source model incurs additional costs that scale, unsurprisingly, on the amount of compute necessary. While this is increasingly easy with coding tools, it can become onerous for enterprises who want to stay on the leading edge as they must re-tune open-weight models as new versions of the base model are released. We believe enterprise readiness and in-house talent are key bottlenecks, as less tech-enabled companies don't have the resources to do this and will likely use an API for faster time to value.
2. **Potential Security Risks.** Bad actors can train the safety alignment — the process that instructs AI models to refuse harmful requests — out of open-weights models, posing risks for enterprise adopters. Once released, weights are beyond the developer's control and modifications are hard to change. We've seen examples of how this can be utilized — in April, the [Financial Times and Alice](#) (AI safety group) used Heretic to remove safety guards from open-weights models from Meta and Google in minutes. On the flip side, some argue that operating with a small number of closed models creates more risk or a "single point of failure."
3. **Liability Attribution Issues for Enterprises.** When companies access closed-source, commercial models from players like OpenAI and Anthropic, they receive IP indemnities that protect them from copyright infringement claims arising from using that model provider (conditional on safety guardrails and correct usage). However, this is not the case with open-weights models, which could create risk for enterprises.
4. **Geopolitical Risk.** Enterprises care about business continuity. Potential bans of Chinese open-weights models in the US, as well as frontier models in China create risk for global businesses. This could cause companies in the US to avoid Chinese open-weights models and opt for US alternatives.

So, where and how are open-weights models being used for today? Primarily smaller deployments requiring speed, customization, or frequent calls. The use cases we hear enterprises using open-weights models for include: 1) coding, 2) document parsing, and 3) other smaller, specialized use cases. These are more isolated use cases, but we believe this can change as open-weights models proliferate. Today, we believe most companies either 1) use a standard enterprise offering from a model provider, or 2) partner directly with a frontier-model provider to customize those models. This allows companies to stay on the frontier and eliminate unnecessary engineering overhead.

As mentioned above, companies have full flexibility over where they run open-weights models – in the cloud, on-prem, or through an API. For companies using open-weights models for data security reasons, one option is to run those workloads on-prem, requiring them to buy or lease GPU clusters. Research from [Carnegie Mellon](#) indicates that payback periods for open-weights models vastly ranges by deployment size, with payback periods ranging from up to 3 months for small deployments (<30B parameter model, used for customer support, document analysis, etc.) to up to 6 years for large deployments (200B-1T parameter model, used for advanced research or enterprise-wide generative applications). This illustrates how open-weights models are not necessarily "free," depending on what an enterprise is using them for.

We believe orchestration and observability will become increasingly important, as open-weights models proliferate. Running LLMs in production requires a broader software and infrastructure stack, regardless of whether an enterprise uses closed or open-weight models. The importance of these layers depends less on the model's licensing structure and more on the number of models and workflow complexity. A few key layers include **gateway layer** (provides a standardized access point to models and manages authentication, rate limits, logging, retries, and provider failover), **routing layer** (selects appropriate model for a given request, given cost, capability, latency, etc.), **orchestration layer** (coordinates multi-step reasoning, turns requests into a series of model/tool calls), and **observability layer** (tracks token usage, evals, quality, latency, and cost), among others. A shift toward open-weight models would likely increase both the number of models enterprises deploy and workload complexity – making routing, orchestration, observability, security, and governance increasingly important parts of the enterprise AI stack.

We don't believe open-weights reflects close to the majority of enterprise AI model spend, but how this trends will be important to watch. In addition to the fine-tuning required to stay at the frontier, switching costs, brand/trust, and aversion to foreign models are also key adoption hurdles. We also believe there are broader misconceptions with open-weights models around data privacy; once a user downloads an reliable open-source model, the model providers lose any insight into the users introduced training data or modified model weights unless that training is done through the provider's hosted API (in which case, uploaded data and information is in the possession of the provider, who may or may not use that data, keep it isolated from other customers, or encrypt it). Protecting one's data becomes an issue of cybersecurity or potential distillation attacks. Overall, shifts of enterprise model spend are subject to change as we see greater model efficiency, developments in open-weights models, and enterprise purchasing behavior.

3 States of the World

Given it's still yet to be seen how the mix of open and closed weights model trends for enterprises, we believe it's useful to **frame three potential scenarios** we see playing out over the next few years:

1. Closed-Weights Models Win. Frontier-model performance remains difficult to replicate, which leads to a small number of well-capitalized labs winning. Enterprises prioritize accuracy, reliability, security, indemnification and ease of deployment over model control or transparency, and they choose to partner with the labs rather than fine-tune open-weights models in-house. AI workloads remain concentrated in hyperscaler clouds, and training demand remains highly compute-intensive. Cost per token falls as models become more efficient, but the cost of compute remains *higher* than it is in the open-weight scenario given the market oligopoly. Security software is a key winner, but more of the security burden is handled by the frontier labs vs. on-prem. Further, the massive electricity demand of training superclusters further exacerbate the demand for reliable, GW scale power: bullish for onsite power providers.

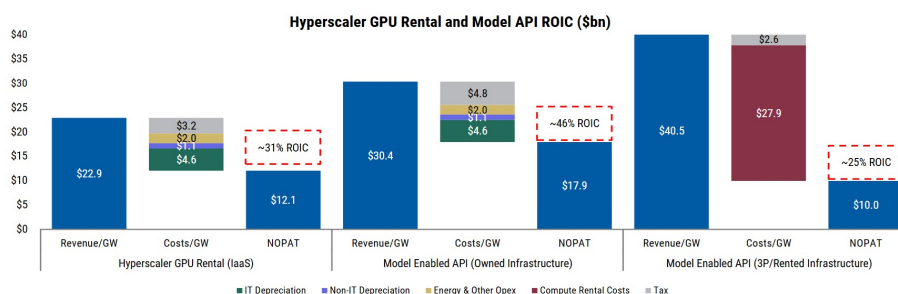
We think the largest beneficiaries in this scenario are:

- **Cloud Providers:** GOOGL, AMZN
- **Model Providers:** GOOGL, META
- **Security Software:** PANW, CRWD, ZS, NTSK, OKTA
- **Optical/Networking:** ANET, LITE, COHR
- **On-Site Power Providers:** BE, INIO, SEI, WMB, LBRT
- **Semis:** NVDA, AVGO

For Meta Marshall's thoughts on implications for security software/networking, see [link](#).

In Scenario 1 (where closed models win) GOOGL's position of strength will be determined by which closed models win. If Gemini returns to the frontier (Gemini 4 is the next likely model for this), we view GOOGL as a primary beneficiary given the ~45% ROIC on running a model API on owned & operated infrastructure. If Scenario 1 plays out and Gemini is not at the frontier, GOOGL's position will remain strong, but it will operate moreso as an infrastructure provider, where we see ~30% ROIC. For more on our ROIC frameworks, please see [The Paths to 25-50% GenAI ROIC \(27 Jul 2026\)](#).

Exhibit 3: We see attractive incremental unit economics and ROIC (25%+) on the three main business models set to emerge through the age of AI inference.



Source: Company data, Morgan Stanley Research Estimates. Note the "ROIC" on a Model-Enabled API running on 3P infrastructure is illustrative and represents revenue, adjusted for training capacity (training costs typically expensed in opex) less the rental cost of compute (COGS), taxed at 21%.

2. Hybrid State of the World. In this scenario, no single model architecture wins – closed frontier models dominate the most complex reasoning and agentic workloads, while open-

weight and smaller models handle high-volume, cost-sensitive or domain-specific tasks. Some workloads will use both, routing tasks to each based on capability and cost. Closed models retain pricing power, but competition from open alternatives limits excess returns. AI deployment spans public cloud, private cloud, on-premise infrastructure and edge devices. Model orchestration, routing, evaluation, and governance, benefitting software players in these areas. Security software becomes increasingly more important as more workloads are handled on-prem or in a more distributed manner.

We think the largest beneficiaries in this scenario are:

- **Cloud Providers:** AMZN, GOOGL, MSFT
- **Infrastructure Software:** DDOG, PLTR, APPN
- **Security Software:** PANW, CRWD, FTNT, ZS, NTSK, OKTA, SAIL, VRNS
- **SaaS:** SAP, NOW
- **Optical/Networking:** CSCO, FFIV
- **On-Site Power Providers:** BE, INIO, SEI, WMB, LBRT
- **Semis:** NVDA

3. Open-Weights Models Win. Open-weight models converge with frontier-level performance and base-model intelligence becomes widely available. Model API prices fall, leading to greater adoption across enterprises, in-line with Jevons Paradox. Companies customize models using proprietary data and run them wherever economics, latency or regulation are most attractive. Data sovereignty, privacy and avoidance of vendor lock-in accelerate adoption by governments and large enterprises. Innovation shifts from pre-training toward fine-tuning, inference optimization, agents, tools and application-specific deployment. AI becomes more decentralized, with substantial growth in private data centers, sovereign clouds, on-prem systems and edge devices. This also drives greater demand for security software to secure on-prem workloads, as well as observability and routing offerings.

We think the largest beneficiaries in this scenario are:

- **Cloud Providers:** MSFT
- **Model Providers:** MiniMax, Knowledge Atlas, BABA, Tencent
- **Infrastructure Software:** PLTR
- **Security Software:** PANW, CRWD, FTNT, OKTA, SAIL
- **SaaS:** SAP, NOW, SHOP
- **On-Prem Infrastructure:** DELL, HPE, NTAP, P
- **Edge Devices:** DELL, HPQ, AAPL
- **VARs/Distributors:** SNX, INGM, CDW
- **On-Site Power Providers:** BE, INIO, SEI, WMB, LBRT
- **Semis:** NVDA

Exhibit 4: Stocks we like across our "3 States of the World" — where closed models win, hybrid wins, and open models win.

	Scenario 1 (Closed Models Win)	Scenario 2 (Hybrid)	Scenario 3 (Open Models Win)
Biggest Beneficiaries	Cloud Providers: GOOGL, AMZN Model Providers: GOOGL, META Security Software: PANW, CRWD, ZS, NTSK, OKTA Optical/Networking: ANET, LITE, COHR On-Site Power Providers: BE, INIO, SEI, WMB, LBRT Semis: NVDA, AVGO	Cloud Providers: AMZN, GOOGL, MSFT Infrastructure Software: DDOG, PLTR, APPN Security Software: PANW, CRWD, FTNT, ZS, NTSK, OKTA, SAIL, VRNS SaaS: SAP, NOW Optical/Networking: CSCO, FFIV On-Site Power Providers: BE, INIO, SEI, WMB, LBRT Semis: NVDA	Cloud Providers: MSFT Model Providers: MiniMax, Knowledge Atlas, BABA, Tencent Infrastructure Software: PLTR Security Software: PANW, CRWD, FTNT, OKTA, SAIL SaaS: SAP, NOW, SHOP On-Prem Infrastructure: DELL, HPE, NTAP, P Edge Devices: DELL, HPQ, AAPL VARs/Distributors: SNX, INGM, CDW On-Site Power Providers: BE, INIO, SEI, WMB, LBRT Semis: NVDA
Other Beneficiaries	Cloud Providers: MSFT Neoclouds: CRWV, NBIS Security Software: SAIL	Model Providers: MiniMax, Knowledge Atlas, BABA, Tencent Neoclouds: CRWV, NBIS SaaS: SHOP, WDAY, CRM, INTU On-Prem Infrastructure: DELL, HPE, NTAP, P Edge Devices: DELL, HPQ, AAPL VARs/Distributors: SNX, INGM, CDW	Cloud Providers: AMZN, GOOGL Neoclouds: CRWV, NBIS SaaS: WDAY, CRM, INTU

Source: Morgan Stanley Research

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Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of AES Corp., Aeva Technologies Inc, Airbnb Inc, Akamai Technologies, Inc.,

Alibaba Group Holding, Alphabet Inc., Amazon.com Inc, American Electric Power Co, Baker Hughes Co, Bloom Energy Corp., Broadcom Inc., Capgemini, Cerebras Systems, Check Point Software Technologies Ltd., Compass, Inc., Constellation Energy Corporation, CoreWeave, Dell Technologies Inc., DigitalOcean Holdings Inc, eBay Inc, Enbridge, Energy Transfer LP, Enterprise Products LP, Figma Inc, Fluence Energy Inc, GE Vernova, GlobalFoundries Inc, Ingram Micro, Innio N.V., Intel Corporation, Intuit, Kanzhun Ltd, Liffort Mobile Inc., Lumentum Holdings Inc, Match Group Inc, Meta Platforms Inc, MGE Energy, Inc., MPLX LP, Nabors Industries Inc., Navan Inc, Nebius Group NV, Netskope, Inc., NextEra Energy Inc, NRG Energy Inc, NVIDIA Corp., NXP Semiconductor NV, ON Semiconductor Corp., Oneok Inc., Perimeter Solutions, Plains All American Pipeline LP, Plains GP Holdings, L.P., Public Service Enterprise Group Inc, Quantinuum, Salesforce, Inc., SAP SE, Semtech Corp., ServiceNow Inc, Solaris Energy Infrastructure, Sunrun Inc, Talen Energy Corp, Targa Resources Corp., TC Energy Corp, Tencent Holdings Ltd., Transocean Ltd., Uber Technologies Inc, Via Transportation Inc, Vistra Corp, WaterBridge Infrastructure LLC, Williams Companies Inc, Wix.Com Ltd, X-Energy Inc..

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from Advanced Micro Devices, Aeva Technologies Inc, Airbnb Inc, Akamai Technologies, Inc., Alibaba Group Holding, Allegro Microsystems Inc, Alphabet Inc., Amazon.com Inc, American Electric Power Co, Amkor Technology Inc, Analog Devices Inc., Array Technologies Inc, Autodesk, Bilibili Inc, Blackline Inc, Bloom Energy Corp., Broadcom Inc., Brookfield Infrastructure Partners L.P., Capgemini, CCC Intelligent Solutions Holdings Inc, CDW Corporation, Cerebras Systems, Check Point Software Technologies Ltd., Chewy Inc, Coherent Corp, CoreWeave, Dassault Systemes SA, Dell Technologies Inc., DigitalOcean Holdings Inc, eBay Inc, Enbridge, Energy Transfer LP, Enterprise Products LP, EVgo Inc, Figma Inc, GE Vernova, Halliburton Co, Hexagon AB, HubSpot, Inc., IBM, Indra, Innio N.V., Instacart, Intel Corporation, Intuit, IonQ Inc, Kinder Morgan Inc, Liberty Energy, Liffort Mobile Inc., Lumentum Holdings Inc, Match Group Inc, Meituan, Meta Platforms Inc, MGE Energy, Inc., Micron Technology Inc, Microsoft, Motorola Solutions Inc, MPLX LP, Nabors Industries Inc., Navan Inc, Nebius Group NV, Netskope, Inc., NextEra Energy Inc, NICE Ltd., NRG Energy Inc, NVIDIA Corp., NXP Semiconductor NV, ON Semiconductor Corp., Oneok Inc., Perimeter Solutions, Plains All American Pipeline LP, Plains GP Holdings, L.P., Public Service Enterprise Group Inc, Quantinuum, RingCentral Inc, Salesforce, Inc., SAP SE, Schlumberger NV, Seagate Technology, Semtech Corp., ServiceNow Inc, Solaris Energy Infrastructure, Sunrun Inc, Talen Energy Corp, Targa Resources Corp., TC Energy Corp, Tencent Holdings Ltd., Texas Instruments, Transocean Ltd., Uber Technologies Inc, Via Transportation Inc, Vistra Corp, WaterBridge Infrastructure LLC, Williams Companies Inc, Wix.Com Ltd, Workday Inc, X-Energy Inc., Zebra Technologies Corporation, Zeta Global Holdings Corp.

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Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from Adobe Inc., Advanced Micro Devices, AES Corp., Airbnb Inc, Akamai Technologies, Inc., Alibaba Group Holding, Allegro Microsystems Inc, Alphabet Inc., Amadeus IT Holdings S.A., Amazon.com Inc, Ambarella Inc, American Electric Power Co, Amkor Technology Inc, Amplitude Inc., Analog Devices Inc., Apple, Inc., AppLovin Corp, Array Technologies Inc, Asana Inc, Atlassian Corporation PLC, Autodesk, Axon Enterprise Inc, Baidu Inc, Baker Hughes Co, Bilibili Inc, Blackline Inc, Bloom Energy Corp., Booking Holdings Inc, Box Inc, Broadcom Inc., Brookfield Infrastructure Partners L.P., Cadence Design Systems Inc, Capgemini, CCC Intelligent Solutions Holdings Inc, CDW Corporation, Cerebras Systems, Check Point Software Technologies Ltd., Chewy Inc, Ciena Corporation, Cisco Systems Inc, Cloudflare Inc, Coherent Corp, Constellation Energy Corporation, CoreWeave, Corning Inc, CrowdStrike Holdings Inc, Dassault Systemes SA, Dell Technologies Inc., DigitalOcean Holdings Inc, DocuSign Inc, DoorDash Inc, DT Midstream, Inc., Dynatrace Inc, eBay Inc, Electronic Arts Inc, Enbridge, Energy Transfer LP, Enterprise Products LP, Expedia Inc., Figma Inc, First Solar Inc, Five9 Inc, Fluence Energy Inc, Freshworks Inc, Full Truck Alliance Co. Ltd, GE Vernova, Gen Digital Inc., GitLab Inc, GlobalFoundries Inc, GoPro Inc, Halliburton Co, HBX Group International PLC, Helmerich & Payne Inc, Hewlett Packard Enterprise, Hexagon AB, HP Inc., HubSpot, Inc., HUAY Inc, IBM, Ingram Micro, Innio N.V., Instacart, Intel Corporation, Intuit, JFrog Ltd., JOYY Inc., Kanzhun Ltd, KE Holdings Inc, Kinder Morgan Inc., Kinetik Holdings Inc, Klaviyo, Inc, Liberty Energy, Liffort Mobile Inc., Logitech International SA, Lumentum Holdings Inc, Marvell Technology Group Ltd, Match Group Inc, Meituan, Meta Platforms Inc, Microchip Technology Inc., Micron Technology Inc., Microsoft, monday.com Ltd, MongoDB Inc, Motorola Solutions Inc, MPLX LP, Nabors Industries Inc., NetApp Inc, NetEase, Inc, NextEra Energy Inc, NICE Ltd., NRG Energy Inc, Nutanix Inc, NVIDIA Corp., NXP Semiconductor NV, ON Semiconductor Corp., Oneok Inc., Oracle Corporation, Palantir Technologies Inc., Palo Alto Networks Inc, Peloton Interactive, Inc., Perimeter Solutions, Pinterest Inc, Plains All American Pipeline LP, Plains GP Holdings, L.P., Playtika Holding Corp, ProFrac Holding Corp, Public Service Enterprise Group Inc, Qorvo Inc, Qualcomm Inc., Reddit Inc, Resideo Technologies Inc, Revolve Group Inc, RingCentral Inc, SailPoint Inc, Salesforce, Inc., SAP SE, Schlumberger NV, Seagate Technology, Semtech Corp., SentinelOne, Inc., ServiceNow Inc, ServiceTitan Inc, Shoals Technologies Group, Silicon Laboratories Inc., Snap Inc., Snowflake Inc., Sonos Inc., Sunrun Inc, Synopsys Inc., Take-Two Interactive Software, Talen Energy Corp, Targa Resources Corp., TC Energy Corp, TD Synnex Corporation, Tenable Holdings Inc, Tenaris SA, Tencent Holdings Ltd., Texas Instruments, Tieto Oyj, Toast, Inc., Tongcheng Travel Holdings, Transocean Ltd., Trip.com Group Ltd, Uber Technologies Inc, UiPath Inc, Unity Software Inc, Varonis Systems, Inc., Vipshop Holdings Ltd, Vistra Corp, WaterBridge Infrastructure LLC, Weibo Corp, Western Digital, Western Midstream Partners LP, Williams Companies Inc, Workday Inc, WW International Inc, Zebra Technologies

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Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: Adobe Inc., Advanced Micro Devices, AES Corp., Aeva Technologies Inc, Airbnb Inc, Akamai Technologies, Inc., Alibaba Group Holding, Allegro Microsystems Inc, Alphabet Inc., Amadeus IT Holdings S.A., Amazon.com Inc, Ambarella Inc, American Electric Power Co, Amkor Technology Inc, Amplitude Inc., Analog Devices Inc., Antero Midstream Corp, Appian Corp, Apple, Inc., AppLovin Corp, Arista Networks, Arm Holdings plc, Array Technologies Inc, Asana Inc, Astera Labs Inc, Atlassian Corporation PLC, Autodesk, Autohome Inc, Axon Enterprise Inc, Baidu Inc, Baker Hughes Co, Bilibili Inc, BILL Holdings Inc, Blackline Inc, Bloom Energy Corp., Booking Holdings Inc, Box Inc, Broadcom Inc., Brookfield Infrastructure Corp, Brookfield Infrastructure Partners L.P., Bumble Inc., C3.ai, Cadence Design Systems Inc, Capgemini, CCC Intelligent Solutions Holdings Inc, CDW Corporation, Cerebras Systems, Check Point Software Technologies Ltd., Chewy Inc, Ciena Corporation, Cisco Systems Inc, Cloudflare Inc, Coherent Corp, Compass, Inc., Constellation Energy Corporation, CoreWeave, Corning Inc, Criteo SA, CrowdStrike Holdings Inc, Dassault Systemes SA, Datadog, Inc., Dell Technologies Inc., Descartes Systems Group Inc, DigitalOcean Holdings Inc, DocuSign Inc, DoorDash Inc, DoubleVerify Holdings Inc, DT Midstream, Inc., Duolingo Inc, eBay Inc, Elastic NV, Electronic Arts Inc, Enbridge, Energy Transfer LP, Enterprise Products LP, Etsy Inc, Everpure, Inc., EVgo Inc, Expedia Inc., F5 Inc, Figma Inc, FIGS, Inc., First Solar Inc, Five9 Inc, Fluence Energy Inc, Fortinet Inc., Freshworks Inc, Full Truck Alliance Co. Ltd, Garmin Ltd, GE Vernova, Gen Digital Inc., GitLab Inc, GlobalFoundries Inc, GoDaddy Inc, GoPro Inc, Halliburton Co, Helmerich & Payne Inc, Hewlett Packard Enterprise, Hexagon AB, HP Inc., HubSpot, Inc., IBM, Indra, Ingram Micro, Innio N.V., Instacart, Intel Corporation, Intuit, IONOS Group SE, IonQ Inc, IQIYI Inc, JD.com, Inc., JFrog Ltd., JOYY Inc, Kanzhun Ltd, KE Holdings Inc, Keysight Technologies Inc, Kinder Morgan Inc, Kinetik Holdings Inc, Klaviyo, Inc, Kornit Digital Ltd., Kuaihou Technology, Liberty Energy, Liftoff Mobile Inc, Lightspeed Commerce Inc., Liveramp Holdings Inc, Logitech International SA, Lumentum Holdings Inc, Lyft Inc, Marvell Technology Group Ltd, Match Group Inc, Meituan, Meta Platforms Inc, MGE Energy, Inc., Microchip Technology Inc., Micron Technology Inc., Microsoft, MNTN Inc, monday.com Ltd, MongoDB Inc, Motorola Solutions Inc, MPLX LP, Nabors Industries Inc., Navan Inc, Navitas Semiconductor Corp, Nebius Group NV, Nemetschek SE, NetApp Inc, Netcompany Group A/S, NetEase, Inc, Netskope, Inc., NextEra Energy Inc, NICE Ltd., NOV Inc., NRG Energy Inc, Nutanix Inc, NVIDIA Corp., NXP Semiconductor Corp., Okta, Inc., ON Semiconductor Corp., Oneok Inc, Opendoor Technologies Inc, Oracle Corporation, PagerDuty, Inc., Palantir Technologies Inc, Palo Alto Networks Inc, Patterson-UTI Energy, PDD Holdings Inc, Peloton Interactive, Inc., Perimeter Solutions, Pinterest Inc, Plains All American Pipeline LP, Plains GP Holdings, L.P., Playtika Holding Corp, Plug Power Inc., ProFrac Holding Corp, Public Service Enterprise Group Inc, Qualcomm Inc., Qualys Inc, Quantinuum, Rapid7 Inc, Reddit Inc, Resideo Technologies Inc, Revolve Group Inc, RingCentral Inc, Roblox Corporation, Sage, SailPoint Inc, Salesforce, Inc., Samsara Inc, SanDisk Corporation., SAP SE, Schlumberger NV, Seagate Technology, Semtech Corp., SentinelOne, Inc., ServiceNow Inc, ServiceTitan Inc, Shopify Inc, Silicon Laboratories Inc, Skyworks Solutions Inc, SmartRent, Inc., Snap Inc., Snowflake Inc., Solaredge Technologies Inc, Solaris Energy Infrastructure, Sonos Inc., South Bow Corp, Sprinkl'r Inc, Sprout Social Inc, SPS Commerce Inc, Sunrun Inc, Synopsys Inc., Take-Two Interactive Software, Talen Energy Corp, Targa Resources Corp., TC Energy Corp, Temenos Group AG, Tenable Holdings Inc, Tencent Holdings Ltd., Teradata, Texas Instruments, Tieto Oyj, Toast, Inc., Tongcheng Travel Holdings, Trade Desk Inc, Transocean Ltd., Trip.com Group Ltd, Twilio Inc, Uber Technologies Inc, UiPath Inc, Unity Software Inc, Varonis Systems, Inc., Vertex Inc., Via Transportation Inc, Vipshop Holdings Ltd, Vistra Corp, WaterBridge Infrastructure LLC, Webtoon Entertainment Inc, Weibo Corp, Western Digital, Western Midstream Partners LP, Williams Companies Inc, Wix.Com Ltd, Wolfspeed, INC, Workday Inc, X-Energy Inc., Yelp Inc, Zebra Technologies Corporation, Zeta Global Holdings Corp, Zillow Group Inc, Zoom Communications, Zscaler Inc.

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: Adobe Inc., Advanced Micro Devices, AES Corp., Airbnb Inc, Akamai Technologies, Inc., Alibaba Group Holding, Allegro Microsystems Inc, Alphabet Inc., Amadeus IT Holdings S.A., Amazon.com Inc, Ambarella Inc, American Electric Power Co, Amkor Technology Inc, Amplitude Inc., Analog Devices Inc., Apple, Inc., AppLovin Corp, Array Technologies Inc, Asana Inc, Atlassian Corporation PLC, Autodesk, Axon Enterprise Inc, Baidu Inc, Baker Hughes Co, Bilibili Inc, Blackline Inc, Bloom Energy Corp., Booking Holdings Inc, Box Inc, Broadcom Inc., Brookfield Infrastructure Partners L.P., Cadence Design Systems Inc, Capgemini, CCC Intelligent Solutions Holdings Inc, CDW Corporation, Cerebras Systems, Check Point Software Technologies Ltd., Chewy Inc, Ciena Corporation, Cisco Systems Inc, Cloudflare Inc, Coherent Corp, Constellation Energy Corporation, CoreWeave, Corning Inc, CrowdStrike Holdings Inc, Dassault Systemes SA, Datadog, Inc., Dell Technologies Inc., DigitalOcean Holdings Inc, DocuSign Inc, DoorDash Inc, DT Midstream, Inc., Dynatrace Inc, eBay Inc, Electronic Arts Inc, Enbridge, Energy Transfer LP, Enterprise Products LP, Etsy Inc, Everpure, Inc., Expedia Inc., Figma Inc, First Solar Inc, Five9 Inc, Fluence Energy Inc, Fortinet Inc., Freshworks Inc, Full Truck Alliance Co. Ltd, Garmin Ltd, GE Vernova, Gen Digital Inc., GitLab Inc, GlobalFoundries Inc, GoDaddy Inc, GoPro Inc, Halliburton Co, HBX Group International PLC, Helmerich & Payne Inc, Hewlett Packard Enterprise, Hexagon AB, HP Inc., HubSpot, Inc., HUYA Inc, IBM, Ingram Micro, Innio N.V., Instacart, Intel Corporation, Intuit, IonQ Inc, IQIYI Inc, JFrog Ltd., JOYY Inc., Kanzhun Ltd, KE Holdings Inc, Kinder Morgan Inc, Kinetik Holdings Inc, Klaviyo, Inc, Liberty Energy, Liftoff Mobile Inc., Logitech International SA, Lumentum Holdings Inc, Lyft Inc, Marvell Technology Group Ltd, Match Group Inc, Meituan, Meta Platforms Inc, Microchip Technology Inc., Micron Technology Inc., Microsoft, monday.com Ltd, MongoDB Inc, Motorola Solutions Inc, MPLX LP, Nabors Industries Inc., Nebius Group NV, NetApp Inc, NetEase, Inc, NextEra Energy Inc, NICE Ltd., NRG Energy Inc, Nutanix Inc, NVIDIA Corp., NXP Semiconductor NV, ON Semiconductor Corp., Oneok Inc., Opendoor Technologies Inc, Oracle Corporation, PagerDuty, Inc., Palantir Technologies Inc., Palo Alto Networks Inc, Peloton Interactive, Inc., Perimeter Solutions, Pinterest Inc, Plains All American Pipeline LP, Plains GP Holdings, L.P., Playtika Holding Corp, Plug Power Inc., ProFrac Holding Corp, Public Service Enterprise Group Inc, Qorvo Inc, Qualcomm Inc., Qualys Inc, Reddit Inc, Resideo Technologies Inc, Revolve Group Inc, RingCentral Inc, Sage, SailPoint Inc, Salesforce, Inc., SanDisk Corporation., SAP SE, Schlumberger NV, Seagate Technology, Semtech Corp., SentinelOne, Inc., ServiceNow Inc, ServiceTitan Inc, Shoals Technologies Group, Shopify Inc, Silicon Laboratories Inc, Snap Inc., Snowflake Inc., Sonos Inc., Sunrun Inc, Synopsys Inc., Take-Two Interactive Software, Talen Energy Corp, Targa Resources Corp., TC Energy Corp, TD Synnex Corporation, Tenable Holdings Inc, Tenaris SA, Tencent Holdings Ltd., Texas Instruments, Tieto Oyj, Toast, Inc., Tongcheng Travel Holdings, Transocean Ltd., Trip.com Group Ltd, Twilio Inc, Uber Technologies Inc, UiPath Inc, Unity Software Inc, Varonis Systems, Inc., Vipshop Holdings Ltd, Vistra Corp, WaterBridge Infrastructure LLC, Weibo Corp, Western Digital, Western Midstream Partners LP, Williams Companies Inc, Wix.Com Ltd, Wolfspeed, INC, Workday Inc, WW International Inc, Zebra Technologies Corporation, Zeta Global Holdings Corp, Zillow Group Inc, Zoom Communications, Zscaler Inc.

An employee, director or consultant of Morgan Stanley is a director of eBay Inc, Elastic NV, HP Inc., Tenable Holdings Inc. This person is not a research analyst or a member of a research analyst's household.

Morgan Stanley & Co. LLC makes a market in the securities of Aeva Technologies Inc, Ambarella Inc, Amplitude Inc., Antero Midstream Corp, Appian Corp, Array Technologies Inc, Asana Inc, Autodesk, Autohome Inc, Blackline Inc, Box Inc, C3.ai, CCC Intelligent Solutions Holdings Inc, Check Point Software Technologies Ltd., DoubleVerify Holdings Inc, DT Midstream, Inc., Elastic NV, FIGS, Inc., Five9 Inc, Garmin Ltd, Grindr Inc., Helmerich & Payne Inc, IQIYI Inc, JOYY Inc., Liveramp Holdings Inc, Manhattan Associates Inc, MGE Energy, Inc., Motorola Solutions Inc, Nabors Industries Inc., NetEase, Inc, NRG Energy Inc, PagerDuty, Inc., Perimeter Solutions, Playtika Holding Corp, Public Service Enterprise Group Inc, Qualys Inc, Rapid7 Inc, Resideo Technologies Inc, Revolve Group Inc, RingCentral Inc, Shoals Technologies Group, Solaris Energy Infrastructure, Sonos Inc., Sprinkl'r Inc, Sprout Social Inc, SPS Commerce Inc, Talen Energy Corp, Targa Resources Corp., TD Synnex Corporation, Twilio Inc, Varonis Systems, Inc., Vertex Inc., Webtoon Entertainment Inc, Weibo Corp, Yelp Inc, ZoomInfo Technologies Inc.

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STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of July 31, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1529	42%	442	47%	29%	750	43%
Equal-weight/Hold	1582	43%	396	42%	25%	773	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	560	15%	97	10%	17%	217	12%
Total	3,674		936			1741	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

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INDUSTRY COVERAGE: Internet

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/31/2026)
Brian Nowak, CFA		
Airbnb Inc (ABNB.O)	U (12/06/2022)	\$151.52
Alphabet Inc. (GOOGL.O)	O (08/11/2015)	\$356.13
Amazon.com Inc (AMZN.O)	O (04/24/2015)	\$271.58
Booking Holdings Inc (BKNG.O)	O (02/23/2026)	\$192.90
DoorDash Inc (DASH.O)	O (02/21/2024)	\$196.16
Expedia Inc. (EXPE.O)	E (01/09/2019)	\$294.74
Instacart (CART.O)	E (01/29/2024)	\$44.60

Lyft Inc (LYFT.O)	E (10/24/2019)	\$15.86
Meta Platforms Inc (META.O)	O (03/20/2023)	\$556.71
Pinterest Inc (PINS.N)	O (07/20/2025)	\$24.01
Reddit Inc (RDDT.N)	O (12/08/2024)	\$140.67
Snap Inc. (SNAP.N)	E (07/22/2024)	\$4.69
Uber Technologies Inc (UBER.N)	++	\$70.36

Matthew Cost

AppLovin Corp (APP.O)	O (04/10/2025)	\$395.90
Compass, Inc. (COMP.N)	E (01/12/2026)	\$11.39
Criteo SA (CRTO.O)	E (01/26/2016)	\$21.82
DoubleVerify Holdings Inc (DV.N)	E (06/25/2024)	\$11.25
Electronic Arts Inc (EA.O)	E (08/04/2021)	\$209.86
Liftoff Mobile Inc. (LFTO.O)	E (06/29/2026)	\$22.20
MNTN Inc (MNTN.N)	E (06/16/2025)	\$9.22
Opendoor Technologies Inc (OPEN.O)	E (07/24/2023)	\$3.77
Playtika Holding Corp (PLTK.O)	E (11/27/2022)	\$3.99
Roblox Corporation (RBLX.N)	O (11/04/2024)	\$35.60
Take-Two Interactive Software (TTWO.O)	O (02/01/2018)	\$242.92
Trade Desk Inc (TTD.O)	E (09/10/2025)	\$18.04
Unity Software Inc (U.N)	O (09/02/2024)	\$31.71
Webtoon Entertainment Inc (WBTN.O)	E (07/22/2024)	\$9.12
Yelp Inc (YELP.N)	U (01/10/2019)	\$26.43
Zillow Group Inc (Z.O)	E (04/18/2018)	\$34.06

Nathan Feather

Bumble Inc. (BMBL.O)	E (03/08/2021)	\$2.85
Chewy Inc (CHWY.N)	O (10/31/2023)	\$22.60
Duolingo Inc (DUOL.O)	E (02/27/2026)	\$134.81
eBay Inc (EBAY.O)	O (04/18/2024)	\$114.01
Etsy Inc (ETSY.N)	E (07/20/2025)	\$81.68
FIGS, Inc. (FIGS.N)	E (02/29/2024)	\$10.70
Grindr Inc. (GRND.N)	O (07/01/2026)	\$17.34
Match Group Inc (MTCH.O)	E (04/18/2024)	\$39.41
Peloton Interactive, Inc. (PTON.O)	E (03/14/2022)	\$6.39
Revolve Group Inc (RVLV.N)	E (10/20/2024)	\$25.11
WW International Inc (WW.O)	E (08/01/2025)	\$14.36

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.

INDUSTRY COVERAGE: Software

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/31/2026)
Adam Wood		
Adobe Inc. (ADBE.O)	U (07/21/2026)	\$250.41
Atlassian Corporation PLC (TEAM.O)	O (01/13/2020)	\$101.02
Intuit (INTU.O)	E (07/21/2026)	\$316.07
Microsoft (MSFT.O)	O (07/21/2026)	\$464.72
Salesforce, Inc. (CRM.N)	E (07/21/2026)	\$184.02
Samsara Inc (IOT.N)	E (03/23/2023)	\$37.27
ServiceNow Inc (NOW.N)	O (09/24/2025)	\$111.23
Shopify Inc (SHOP.O)	O (04/19/2024)	\$117.15
Workday Inc (WDAY.O)	U (07/21/2026)	\$160.34
Chris Quintero		
BILL Holdings Inc (BILL.N)	E (06/10/2025)	\$45.13
Blackline Inc (BL.O)	E (07/21/2026)	\$31.40
Box Inc (BOX.N)	E (05/21/2024)	\$31.55

CCC Intelligent Solutions Holdings Inc (CCC.O)	O (11/13/2024)	\$6.09
Descartes Systems Group Inc (DSGX.O)	O (01/15/2026)	\$75.16
DocuSign Inc (DOCU.O)	E (01/16/2024)	\$54.83
Manhattan Associates Inc. (MANH.O)	E (10/21/2025)	\$191.36
Navan Inc (NAVN.O)	O (11/24/2025)	\$26.62
ServiceTitan Inc (TTAN.O)	O (01/20/2026)	\$83.06
SPS Commerce Inc (SPSC.O)	U (07/21/2026)	\$73.39
Vertex Inc. (VERX.O)	E (07/21/2026)	\$12.91
Via Transportation Inc (VIA.N)	O (01/20/2026)	\$20.34

Elizabeth Porter, CFA

Amplitude Inc. (AMPL.O)	O (01/15/2026)	\$8.88
Asana Inc (ASAN.N)	U (05/20/2025)	\$8.32
Autodesk (ADSK.O)	O (08/23/2024)	\$234.20
Figma Inc (FIG.N)	E (08/25/2025)	\$24.32
Five9 Inc (FIVN.O)	E (10/10/2022)	\$27.51
Freshworks Inc (FRSH.O)	E (10/18/2021)	\$11.36
GoDaddy Inc (GDDY.N)	E (07/19/2021)	\$82.74
HubSpot, Inc. (HUBS.N)	O (03/21/2023)	\$237.36
Klaviyo, Inc (KVYO.N)	O (04/29/2026)	\$17.84
Liveramp Holdings Inc (RAMP.N)	E (01/13/2025)	\$37.87
monday.com Ltd (MNDY.O)	O (08/12/2025)	\$87.15
NICE Ltd. (NICE.O)	E (07/21/2026)	\$98.79
RingCentral Inc (RNG.N)	E (08/08/2023)	\$55.62
Sprinklr Inc (CXM.N)	E (07/19/2021)	\$6.36
Sprout Social Inc (SPT.O)	E (11/17/2020)	\$8.58
Toast, Inc. (TOST.N)	O (12/16/2021)	\$32.27
Twilio Inc (TWLO.N)	O (02/24/2025)	\$197.35
Wix.Com Ltd (WIX.O)	E (07/21/2026)	\$55.03
Zeta Global Holdings Corp (ZETA.N)	E (08/01/2024)	\$21.60
Zoom Communications (ZM.O)	E (10/11/2022)	\$96.07
ZoomInfo Technologies Inc (GTM.O)	E (02/01/2024)	\$3.30

Meryl R Thomas, CFA

Lightspeed Commerce Inc. (LSPD.N)	E (02/18/2021)	\$10.11
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Meta A Marshall

Check Point Software Technologies Ltd. (CHKP.O)	E (10/16/2023)	\$127.13
CrowdStrike Holdings Inc (CRWD.O)	O (03/10/2026)	\$190.86
Fortinet Inc. (FTNT.O)	E (07/21/2026)	\$161.95
Gen Digital Inc. (GEN.O)	E (06/07/2024)	\$27.45
Netskope, Inc. (NTSK.O)	O (10/13/2025)	\$12.59
Okta, Inc. (OKTA.O)	O (12/02/2024)	\$141.93
Palo Alto Networks Inc (PANW.O)	O (10/10/2017)	\$331.83
Qualys Inc (QLYS.O)	U (02/09/2021)	\$144.75
Rapid7 Inc (RPD.O)	U (07/21/2026)	\$9.65
SailPoint Inc (SAIL.O)	O (09/02/2025)	\$16.77
SentinelOne, Inc. (S.N)	E (12/02/2024)	\$19.06
Tenable Holdings Inc (TENB.O)	E (12/02/2024)	\$32.64
Varonis Systems, Inc. (VRNS.O)	E (01/26/2026)	\$39.95
Zscaler Inc (ZS.O)	E (04/22/2026)	\$151.20

Ryan Lountzis

CoreWeave (CRWV.O)	E (04/22/2025)	\$71.77
Nebius Group NV (NBIS.O)	E (01/15/2026)	\$190.41

Sanjit K Singh

Akamai Technologies, Inc. (AKAM.O)	O (01/12/2026)	\$115.18
Appian Corp (APPN.O)	E (04/30/2026)	\$26.96

C3.ai (AI.N)	U (01/04/2021)	\$9.18
Cloudflare Inc (NET.N)	O (12/02/2024)	\$278.98
Datadog, Inc. (DDOG.O)	O (01/12/2026)	\$267.97
DigitalOcean Holdings Inc (DOCN.N)	O (01/16/2025)	\$117.50
Dynatrace Inc (DT.N)	E (02/13/2024)	\$44.32
Elastic NV (ESTC.N)	E (07/21/2026)	\$65.79
GitLab Inc (GTLB.O)	E (01/12/2026)	\$34.51
JFrog Ltd. (FROG.O)	E (07/21/2026)	\$79.79
MongoDB Inc (MDB.O)	O (04/12/2023)	\$337.48
Oracle Corporation (ORCL.N)	E (01/15/2019)	\$129.87
PagerDuty, Inc. (PD.N)	U (07/21/2026)	\$10.58
Palantir Technologies Inc. (PLTR.O)	E (02/04/2025)	\$123.06
Snowflake Inc. (SNOW.N)	O (06/24/2025)	\$293.28
UiPath Inc (PATH.N)	E (09/07/2022)	\$12.76

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* Historical prices are not split adjusted.

INDUSTRY COVERAGE: IT Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/31/2026)
Erik W Woodring		
Apple, Inc. (AAPL.O)	O (05/26/2009)	\$308.91
CDW Corporation (CDW.O)	O (06/23/2026)	\$147.81
Cricut Inc (CRCT.O)	U (08/12/2021)	\$4.69
Dell Technologies Inc. (DELL.N)	E (06/01/2026)	\$405.37
Everpure, Inc. (P.N)	E (06/11/2024)	\$77.17
Garmin Ltd (GRMN.N)	E (02/18/2026)	\$293.78
GoPro Inc (GPRO.O)	U (12/12/2023)	\$0.75
Hewlett Packard Enterprise (HPE.N)	E (11/16/2025)	\$47.90
HP Inc. (HPQ.N)	U (11/16/2025)	\$27.27
IBM (IBM.N)	E (01/18/2023)	\$223.65
Ingram Micro (INGM.N)	E (06/11/2025)	\$28.21
Kornit Digital Ltd. (KRNT.O)	E (11/06/2025)	\$15.22
Logitech International SA (LOGI.O)	U (01/20/2026)	\$102.48
NetApp Inc (NTAP.O)	U (01/20/2026)	\$178.50
Resideo Technologies Inc (REZI.N)	O (08/11/2025)	\$34.30
Seagate Technology (STX.O)	O (03/26/2024)	\$856.13
SmartRent, Inc. (SMRT.N)	++	\$1.01
Sonos Inc. (SONO.O)	E (11/06/2025)	\$14.66
TD Synnex Corporation (SNX.N)	O (06/11/2025)	\$255.70
Teradata (TDC.N)	O (04/08/2025)	\$30.99
Western Digital (WDC.O)	O (04/16/2025)	\$544.84
Sanjit K Singh		
Nutanix Inc (NTNX.O)	E (01/12/2026)	\$59.01

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INDUSTRY COVERAGE: Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/31/2026)
Joseph Moore		
Advanced Micro Devices (AMD.O)	E (06/09/2024)	\$476.15
Aeva Technologies Inc (AEVA.O)	E (07/19/2021)	\$17.54
Allegro Microsystems Inc (ALGM.O)	O (02/13/2026)	\$41.50
Ambarella Inc (AMBA.O)	O (03/29/2016)	\$86.00

Amkor Technology Inc (AMKR.O)	E (11/08/2023)	\$49.87
Analog Devices Inc. (ADI.O)	O (11/16/2023)	\$367.41
Astera Labs Inc (ALAB.O)	O (05/11/2025)	\$311.23
Broadcom Inc. (AVGO.O)	O (06/09/2024)	\$389.28
Cerebras Systems (CBR.S.O)	O (06/08/2026)	\$198.71
GlobalFoundries Inc (GFS.O)	E (10/28/2024)	\$49.99
Intel Corporation (INTC.O)	E (02/22/2023)	\$90.20
IonQ Inc (IONQ.N)	E (04/25/2023)	\$36.44
Marvell Technology Group Ltd (MRVL.O)	E (09/14/2015)	\$187.56
Microchip Technology Inc. (MCHP.O)	E (07/10/2024)	\$74.29
Micron Technology Inc. (MU.O)	O (10/06/2025)	\$823.03
Navitas Semiconductor Corp (NVTS.O)	U (04/06/2025)	\$10.86
NVIDIA Corp. (NVDA.O)	O (03/16/2023)	\$200.75
NXP Semiconductor NV (NXPI.O)	O (02/11/2025)	\$229.16
ON Semiconductor Corp. (ON.O)	++	\$81.61
Qorvo Inc (QRVO.O)	E (10/28/2025)	\$90.57
Qualcomm Inc. (QCOM.O)	E (06/24/2026)	\$147.61
Quantinuum (QNT.O)	E (06/29/2026)	\$51.41
SanDisk Corporation. (SNDK.O)	O (03/03/2025)	\$1,214.83
Semtech Corp. (SMTC.O)	E (04/06/2025)	\$117.82
Silicon Laboratories Inc. (SLAB.O)	E (01/19/2021)	\$217.50
Skyworks Solutions Inc (SWKS.O)	E (11/28/2018)	\$62.28
Texas Instruments (TXN.O)	U (04/13/2020)	\$275.74
Wolfspeed, INC (WOLF.N)	NR (04/06/2025)	\$23.62
Lee Simpson		
Arm Holdings plc (ARM.O)	E (04/07/2026)	\$239.69
Cadence Design Systems Inc (CDNS.O)	O (02/14/2024)	\$340.02
Synopsys Inc. (SNPS.O)	E (02/27/2026)	\$388.76

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INDUSTRY COVERAGE: China Internet and Other Services

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/31/2026)
Eddy Wang, CFA		
Autohome Inc (ATHM.N)	E (02/09/2023)	\$23.22
Full Truck Alliance Co. Ltd (YMM.N)	O (07/05/2023)	\$9.49
JD.com, Inc. (JD.O)	U (11/10/2025)	\$33.01
Kanzhun Ltd (BZ.O)	O (08/04/2021)	\$16.28
KE Holdings Inc (BEKE.N)	O (03/16/2022)	\$16.95
PDD Holdings Inc (PDD.O)	O (03/02/2023)	\$88.56
Vipshop Holdings Ltd (VIPS.N)	E (02/24/2022)	\$15.34
Gary Yu		
Alibaba Group Holding (BABA.N)	O (02/24/2025)	\$122.25
Baidu Inc (BIDU.O)	E (05/17/2024)	\$111.11
Meituan (3690.HK)	O (08/29/2024)	HK\$92.50
Tencent Holdings Ltd. (0700.HK)	O (03/19/2020)	HK\$475.20
Rebecca Xu		
HUYA Inc (HUYA.N)	E (05/16/2024)	\$2.45
IQIYI Inc (IQ.O)	E (01/19/2023)	\$1.27
JOYY Inc. (JOYY.O)	E (06/02/2022)	\$73.57
Weibo Corp (WB.O)	U (05/17/2024)	\$8.15
Yang Liu		
Bilibili Inc (BILI.O)	O (04/13/2026)	\$19.10

Kuaishou Technology (1024.HK)	O (05/26/2026)	HK\$43.70
NetEase, Inc (NTES.O)	O (01/08/2025)	\$133.27
Tongcheng Travel Holdings (0780.HK)	O (01/04/2019)	HK\$14.28
Trip.com Group Ltd (TCOM.O)	O (05/17/2021)	\$47.01

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INDUSTRY COVERAGE: Diversified Utilities / IPPs

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/31/2026)
David Arcaro, CFA		
AES Corp. (AES.N)	E (03/06/2026)	\$14.68
American Electric Power Co (AEP.O)	O (03/10/2020)	\$127.85
Constellation Energy Corporation (CEG.O)	O (03/25/2026)	\$262.75
MGE Energy, Inc. (MGEE.O)	U (11/17/2021)	\$79.53
NextEra Energy Inc (NEE.N)	O (09/06/2022)	\$86.92
NRG Energy Inc (NRG.N)	E (12/09/2022)	\$134.29
Public Service Enterprise Group Inc (PEG.N)	O (07/02/2020)	\$76.68
Solaris Energy Infrastructure (SEI.N)	O (12/02/2025)	\$51.41
Talen Energy Corp (TLN.O)	O (03/12/2025)	\$334.10
Vistra Corp (VST.N)	O (03/25/2019)	\$148.19

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INDUSTRY COVERAGE: Telecom & Networking Equipment

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/31/2026)
Meta A Marshall		
Arista Networks (ANET.N)	O (10/31/2023)	\$180.35
Axon Enterprise Inc (AXON.O)	O (12/03/2024)	\$527.76
Ciena Corporation (CIEN.N)	E (10/10/2025)	\$377.05
Cisco Systems Inc (CSCO.O)	O (04/08/2024)	\$115.99
Coherent Corp (COHR.N)	E (12/13/2023)	\$262.89
Corning Inc (GLW.N)	E (06/13/2024)	\$138.25
F5 Inc (FFIV.O)	E (04/12/2022)	\$402.57
Keysight Technologies Inc (KEYS.N)	O (07/13/2026)	\$319.08
Lumentum Holdings Inc (LITE.O)	E (05/12/2021)	\$713.94
Motorola Solutions Inc (MSI.N)	O (12/17/2025)	\$435.75
Zebra Technologies Corporation (ZBRA.O)	E (12/02/2024)	\$293.82

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INDUSTRY COVERAGE: Midstream Energy Infrastructure

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/31/2026)
Robert S Kad		
Antero Midstream Corp (AM.N)	E (04/21/2026)	\$21.97
Brookfield Infrastructure Corp (BIPC.N)	U (03/22/2026)	\$42.30
Brookfield Infrastructure Partners L.P. (BIP.N)	O (03/22/2026)	\$41.76
DT Midstream, Inc. (DTM.N)	E (04/21/2026)	\$138.00
Enbridge (ENB.TO)	E (10/25/2024)	C\$76.28
Energy Transfer LP (ET.N)	E (12/17/2025)	\$20.36
Enterprise Products LP (EPD.N)	U (12/17/2025)	\$38.05
Hess Midstream LP (HESM.N)	U (06/09/2026)	\$40.83
Kinder Morgan Inc. (KMI.N)	E (09/16/2024)	\$32.18

Kinetik Holdings Inc (KNTK.N)	O (07/29/2026)	\$50.57
MPLX LP (MPLX.N)	E (03/19/2021)	\$58.45
Oneok Inc. (OKE.N)	E (07/29/2026)	\$90.81
Plains All American Pipeline LP (PAA.O)	E (10/25/2024)	\$24.57
Plains GP Holdings, L.P. (PAGP.O)	E (10/25/2024)	\$26.33
South Bow Corp (SOBO.TO)	U (10/25/2024)	C\$51.98
Targa Resources Corp. (TRGP.N)	O (03/19/2021)	\$270.37
TC Energy Corp (TRP.TO)	E (06/09/2026)	C\$94.40
WaterBridge Infrastructure LLC (WBI.N)	O (04/21/2026)	\$34.22
Western Midstream Partners LP (WES.N)	E (06/09/2026)	\$46.57
Williams Companies Inc (WMB.N)	O (10/03/2024)	\$71.54

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INDUSTRY COVERAGE: Technology - Software & Services

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/31/2026)
Adam Wood		
SAP SE (SAP.N)	O (02/20/2026)	\$183.62
SAP SE (SAPG.DE)	O (03/20/2020)	€157.68
George W Webb		
Amadeus IT Holdings S.A. (AMA.MC)	O (04/10/2026)	€53.34
Capgemini (CAPP.PA)	E (02/19/2026)	€102.50
Dassault Systemes SA (DAST.PA)	E (03/17/2026)	€20.05
Hexagon AB (HEXAb.ST)		SKr 93.62
IONOS Group SE (IOSn.DE)	E (05/12/2025)	€31.42
Nemetschek SE (NEKG.DE)	E (07/13/2022)	€58.05
Netcompany Group A/S (NETCG.CO)	U (01/12/2026)	DKr 323.00
Sage (SGE.L)	O (12/08/2021)	973p
Marie-Ange Riggio		
Indra (IDR.MC)	O (07/14/2026)	€56.80
Mark Hyatt		
HBX Group International PLC (HBX.MC)	E (04/17/2026)	€7.72
Temenos Group AG (TEMN.S)	U (12/15/2017)	SFr 66.95
Tieto Oyj (TIETO.HE)	U (01/12/2026)	€18.54

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INDUSTRY COVERAGE: Energy Services & Equipment

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/31/2026)
Joe Laetsch, CFA		
Baker Hughes Co (BKR.O)		\$60.49
Halliburton Co (HAL.N)	O (01/04/2022)	\$32.25
Helmerich & Payne Inc (HP.N)	U (03/26/2025)	\$34.57
Liberty Energy (LBRT.N)	O (03/26/2025)	\$18.71
Nabors Industries Inc. (NBR.N)	O (10/17/2023)	\$85.67
NOV Inc. (NOV.N)	E (12/15/2025)	\$19.43
Patterson-UTI Energy (PTEN.O)	E (10/17/2023)	\$10.48
Perimeter Solutions (PRM.N)	O (06/09/2022)	\$30.61
ProFrac Holding Corp (ACDC.O)	U (10/17/2024)	\$4.04
Schlumberger NV (SLB.N)	O (09/10/2019)	\$49.59
Tenaris SA (TS.N)	E (07/15/2026)	\$57.33
Transocean Ltd. (RIG.N)	E (07/07/2020)	\$5.32

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INDUSTRY COVERAGE: Clean Tech

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/31/2026)
David Arcaro, CFA		
Array Technologies Inc (ARRY.O)	E (10/17/2023)	\$5.22
Bloom Energy Corp. (BE.N)	O (01/09/2023)	\$205.81
Enphase Energy Inc (ENPH.O)	U (04/23/2025)	\$37.54
EVgo Inc (EVGO.O)	E (06/02/2025)	\$1.58
First Solar Inc (FSLR.O)	O (12/08/2023)	\$211.03
Fluence Energy Inc (FLNC.O)	E (11/22/2021)	\$13.93
GE Vernova (GEV.N)	O (08/01/2024)	\$990.29
Innio N.V. (INIO.O)	O (06/29/2026)	\$24.42
Plug Power Inc. (PLUG.O)	U (12/06/2023)	\$2.06
Shoals Technologies Group (SHLS.O)	E (12/16/2025)	\$8.74
Solaredge Technologies Inc (SEDG.O)	E (12/16/2025)	\$41.20
Sunrun Inc (RUN.O)	E (04/23/2025)	\$9.81
X-Energy Inc. (XE.O)	O (05/19/2026)	\$16.31

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